Interrogation Cannot Identify, Intervention Can: Computational Indicator Properties of Consciousness Under Test on Language-Model Agents

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Abstract

Whether consciousness depends on more than computation is usually argued in principle. We report what happens when the computational claims are put to work. Butlin, Long and colleagues distilled consciousness theories into computational indicator properties; we built an executable harness for global-workspace and attention-schema indicators and ran preregistered experiments on language-model agents. First, the functional specifications underdetermine outcomes. The predicted benefit of a capacity-limited broadcast rests on an assumption no theory states, that the consumer is itself capacity-limited; frontier models are not, and the benefit dies under compute matching. Implementing broadcast and recruitment forced two further unstated choices, what triggers revision and who computes salience, and those choices, not the workspace, decided every result: a full workspace loop, an ablated hub, and a flat prompt performed identically on a task with genuine headroom. Second, behavioral identification fails: a prompted imitation out-describes the real architecture on self-report, and the two defensible constraint-probe designs order real and imitation oppositely, an empirical, machine-side counterpart of the unfolding argument. Third, intervention escapes the regress: perturbing the architecture's actual capacity dial, self-reports covary with the dial for the real system (Spearman $\rho = 0.48$ overall, exactly 1.000 on engaged replies) and stay frozen for imitations, across two further indicator vocabularies and a second model, defeated only by leaking the schedule. Behavioral assessment of computational properties is possible as blinded intervention, and what it certifies is causal coupling to the manipulated implementation, not possession of the computation.

1 Introduction

Computationalism holds that consciousness is fixed by computational or functional organization, independent of substrate; implementationism holds that how the computation is physically realized matters essentially. The debate is usually conducted at the level of principle, most sharply in the unfolding argument and its aftermath [2]: if behaviorally equivalent systems can differ in the causal structure a theory deems essential, then input-output evidence cannot decide between them, and theories built on such structure risk empirical vacuity. This paper approaches the same question from below, with experiments. We take the computational claims of two consciousness theories at their word, implement them, and ask three questions that the debate usually leaves to intuition: whether the computational specification fixes what the system does, whether the presence of the specified organization can be identified from behavior at all, and what kind of access turns identification from impossible to routine.

Global workspace theory (GWT) makes a functional claim that is unusual among theories of consciousness in being directly testable in an artificial system: the narrow capacity of conscious access is not a defect that brains work around but the mechanism by which specialized, parallel modules are forced to integrate [1, 3]. If that is right, an agent built around a limited-capacity broadcast channel should, under the right conditions, outperform the same agent with the limit removed.

Butlin, Long and colleagues [4, 5] distilled GWT and several other theories into fourteen computational *indicator properties*, offering the most systematic rubric to date for asking whether an AI system has the architectural features these theories associate with consciousness. The rubric, however, is applied by inspection: one reads a system's documentation and judges whether the property is present. To our knowledge there is no executable test suite that measures indicator properties behaviorally across systems; the closest existing artifact is a single project that scores itself against the list [6].

This paper reports what happened when we built such a suite and used it. We make no claim about consciousness in either direction: every result concerns what the computational specifications determine, and what an assessor can know. The contributions:

1. **An open test harness.** Specialist modules hold facts; a capacity-limited broadcast channel with salience competition is dialed from zero to unlimited; task families are designed in matched pairs so that single variables can be isolated. Oracle models (deterministic readers that use exactly what reaches them) calibrate every design before a paid model runs, and the harness is engineered so that measurement bugs surface as impossibilities rather than as findings (Section 2.1).

2. **The specification underdetermines: capacity.** With required information held complete and constant, removing 48 confusable distractors improves accuracy by $+0.026$ ($p = 0.19$; $n = 1{,}200$ per cell). The bottleneck's predicted value rests on an assumption no theory states, a capacity-limited consumer, and the substrate falsifies it; two silent capacity-dependent confounds nearly produced the predicted effect anyway (Section 3.1).

3. **The specification underdetermines: broadcast and recruitment.** A preregistered implementation of GWT's conflict-resolution functions on a task family with genuine headroom finds that revision under broadcast is ubiquitous rather than conflict-specific, that recruitment collapses or recovers with the choice of salience function alone, and that a full workspace loop, an ablated hub, and a flat prompt decide identically. The outcomes are set by choices the theory never makes: what triggers revision, and who computes salience (Section 3.2).

4. **Interrogation cannot identify the property.** A model that architecturally lacks a limited-capacity workspace can be prompted into passing behavioral tests for one, and the two defensible probe designs order the real and imitation systems *oppositely*: an unfolding-style unidentifiability result, obtained empirically in machines, where the ground truth needed to break the tie is exactly the architectural access behavioral testing was meant to replace (Section 3.3).

5. **Intervention beats interrogation.** Turning the architecture's actual knobs and testing whether self-reports covary with the dial separates the real system from every black-box imitation we could buy, across three indicator vocabularies and two models, at a quarter of interrogation's sample cost and with no content ground truth. An imposter-escalation ladder locates the method's exact boundary: coaching on genuine transcripts buys nothing, while leaking the dial's setting inverts the test. The operative requirement is blinding, rediscovered from first principles (Section 3.4).

Everything below cost \$126.24 of API spend across roughly 75,000 model calls (\$107.68 for Experiments 1 and 3; \$10.56 for the preregistered conflict-and-recruitment run and its rater-salience arm; the four preregistered perturbation grids together cost \$14.24), which we note because the main practical obstacle to this kind of work is not compute but measurement: most of the calls were spent discovering that earlier, cheaper versions of the experiment could not answer the question.

1.1 Related work

Global workspace theory and its implementations. GWT originates with Baars [1] and was developed neuronally by Dehaene and colleagues [3]. Several strands of machine-learning work implement workspace-like bottlenecks: shared-workspace transformers [7], the global latent workspace program [8], and the Conscious Turing Machine [9] with its recent foundation-model implementation [10]. Closest to our capacity experiment, Dossa et al. [11] trained an embodied audiovisual agent designed against the GWT indicator properties and found the workspace architecture beat a recurrent control, with the advantage *largest at small working-memory sizes*; that result is the direct motivation for our capacity sweep. A proposed GWT architecture for LLM multi-agent systems [12] reports no experiments and does not vary its capacity threshold. We are not aware of prior work that sweeps workspace capacity in an LLM agent and measures task performance against matched controls.

Attention schema theory. Wilterson and Graziano [13] showed a reinforcement-learning agent with an attention schema learns attentional control where an agent without one is "comparatively incapacitated," and Piefke et al. [14] sharpened this to a conditional prediction: the schema's benefit scales with the agent's uncertainty about its own attentional state. Our harness implements a predictive, control-enabling schema for its rung-4 architecture, and our prompted-passing experiment uses attention-schema vocabulary as a second indicator.

Self-report and introspection in language models. Chalmers [15] surveys the candidate reasons for denying consciousness in current LLMs. A growing empirical literature asks whether LLM self-reports mean anything: concept-injection experiments provide causal evidence of limited functional introspection [16], while follow-up work argues that models often detect generic perturbation rather than introspect [17], and probing studies find no evidence that small models' denials of sentience are untruthful [18]. Our prompted-passing experiment is a black-box relative of this literature: rather than asking whether reports are causally grounded in internal state, we ask whether a report of an *architectural* property can be distinguished behaviorally from the property itself, and we find that the discriminating probes are the ones that target what the property forbids.

Irrelevant context. Shi et al. [19] found that adversarially constructed irrelevant context degrades LLM arithmetic reasoning, and position effects in long contexts are well documented [20]. We observe the opposite sign under a different manipulation: same-format, easily-rejected factual statements *improve* accuracy monotonically in our task (Section 3.1.4). We flag rather than explain this.

Tractable questions. Comsa [21] argues research should shift from the intractable question of whether AI is conscious toward measurable proxies. This paper is an exercise in that program, and its results suggest the proxies themselves need more adversarial scrutiny than they currently receive.

2 Methods and Materials

2.1 The test harness

2.1.1 Architecture ladder

The harness implements an *architecture ladder*: a sequence of agent designs that differ by exactly one structural feature, driven by the same underlying language model through a one-method interface, so that no rung has privileged access to the model.

- **Rung 0, direct**: every module's content in a single prompt.
- **Rung 1, scratchpad**: serial working memory without modularity.
- **Rung 2, unrestricted multi-agent**: specialist modules with an unlimited broadcast channel (GWT-1 only).
- **Rung 3, workspace**: the experimental system. Modules bid for a broadcast channel with a hard token budget per cycle; salience is task-relevance plus a decay for content already delivered in full; the controller sees only what won (GWT-1 through GWT-3). Rung 2 is literally rung 3 with the budget removed, so their comparison isolates the bottleneck.
- **Rung 4, attention schema**: rung 3 plus a predictive model of the system's own attention that both reports and biases the next competition (AST-1). Following [14], attentional noise is a tunable parameter; in oracle validation the schema buys nothing at zero noise and pays exactly as its own accuracy degrades, reproducing the conditional prediction.

2.1.2 Task families

Integration under confusable distraction. Each of k required containers holds a value ("The silver case contains 82."); the task is to report the sum of exactly the named containers. Every distractor is a *near-twin* of a required container (`pale_red_box` against `red_box`), so rejecting it requires exact name matching rather than a glance. **Matched control arm**: identical count and text volume, but the twins derive from containers *not* on the required list, so rejection is trivial. Both arms share required containers and the correct answer by construction (values are drawn from a separate random stream), so any difference between arms is attributable to confusability alone. **Throughput control**: independent single-module questions with nothing to integrate, on which a capacity limit should only hurt; used to detect the boring confound in which a narrow channel helps merely because shorter prompts are easier.

2.1.3 Oracle calibration and measurement discipline

Every design is validated against *oracle models* before any paid call: deterministic readers that perfectly use whatever information reaches them and nothing else. An oracle's score is therefore a readout of how much task-relevant information the capacity limit admitted, which is the quantity a design manipulates. If a sweep is flat under the oracle, the instrument cannot detect a capacity effect and a flat curve from a real model would be uninterpretable. Oracle sweeps are free, and they caught one bug that would otherwise have been published as a result (Section 2.1.4).

Engineering discipline matters more here than usual, because in this experiment measurement bugs are not noise, they are *mimics*: a silent capacity-dependent failure looks exactly like a capacity effect. The harness is test-driven (322 tests); the capacity limiter's invariant is checked at every budget from 0 to 64 tokens and was mutation-tested (a deliberately introduced one-token leak trips 72 tests); every model response is cached to disk keyed on model, effort, token limit, system prompt and prompt, so any run replays at zero cost; spend is bounded by an atomically reserved call cap (a naive counter overspent by 30% under concurrency in testing); and refusals and truncations raise exceptions rather than returning text, because a truncated answer scores zero and is indistinguishable from the effect under study. Safety-classifier refusals, which we observed to be transient false positives on this arithmetic task, are retried and then excluded and counted, never scored: scoring them zero would concentrate losses in long-prompt conditions and manufacture the predicted result.

2.1.4 Case study: the starvation mimic

An early oracle run showed both experimental arms scoring zero at zero attentional noise while *added noise improved them.* Noise cannot help an oracle, so this was a harness fault, not a finding. The cause: a module truncated at the capacity boundary still appeared in the broadcast text, so the "already delivered" salience decay fired for it; every module then decayed identically each cycle, relative order never changed, and whichever module lost the first competition lost every one after it. Its fact never reached the controller at any number of cycles. The failure was silent, deterministic, and capacity-dependent, that is, shaped exactly like the effect the experiment exists to measure. The fix (only content delivered in full counts as delivered) is guarded by a regression suite whose general invariant is that noise must never improve an oracle's score.

2.2 Models, preregistration, and statistics

All experiments drive the architectures with Anthropic models through a one-method interface (Claude Haiku 4.5 unless stated; replications on Claude Opus 5 and Claude Sonnet 5 where noted). The capacity dose-response, the conflict-and-recruitment experiment, and all five perturbation grids were preregistered in-repository: hypotheses, cells, seeds, exclusion rules, power analyses, and the exact invocation frozen and committed to the public git history before the code that produced any reported number ran, with amendments (all pre-data) committed as dated addenda. Every model response is disk-cached, so every number replays at zero cost. Statistics are two-proportion z tests (Fisher exact below ten per cell), paired McNemar tests for within-scenario architecture contrasts, and Spearman rank correlation with permutation tests for the perturbation grids; alpha 0.05 two-sided throughout.

3 Results

3.1 Experiment 1: does the bottleneck help?

3.1.1 The ceiling problem and the capability-by-difficulty probe

We first ran the sweep on Claude Opus 5 (effort low, thinking enabled) with eight required values and up to 48 confusable distractors. The model scored 1.00 in every condition where the required information was present, including 240 pooled trials with all 48 distractors admitted (Wilson 95% CI $[0.984, 1.000]$), and 0.00 in floor conditions where the information was genuinely absent, confirming it does not fabricate. A ceiling cannot discriminate between hypotheses: there is no degradation for a filter to prevent. A cheap probe (12 trials per condition, $0.11) across model capability and task difficulty located a measurable regime: Claude Haiku 4.5 combining twelve values sits near 0.6 accuracy, clear of both floor and ceiling. A companion sweep pins down which knob opens the window: Claude Haiku 4.5 at *eight* required facts is itself at ceiling (0.92 to 0.98 at every capacity, both arms), so the measurable regime is set by how many facts must be integrated, not by distractor load; quadrupling distractors never broke a ceiling, while adding four required facts did. All subsequent Experiment 1 results use Claude Haiku 4.5 at twelve facts. We note the corollary, which we believe generalizes: *functional claims about capacity limits can only be tested on systems that are actually strained*, and for frontier models that regime is surprisingly hard to reach.

3.1.2 The repetition confound

Our first properly powered contrast compared a filtered condition (capacity 20, which admits exactly the required facts) against a flooded condition (unlimited capacity, all 48 distractors) at 3,000 trials per cell. It showed a bottleneck benefit of +0.036 ($z = 2.85$, $p = 0.0043$). This number is wrong, and the reason is instructive. With no filtering, every module rebroadcasts on every cycle, so widening the channel had also *tripled* how often each required fact appeared in the controller's prompt (audit: 1.0 versus 3.0 appearances; 123 versus 1,275 tokens). Repetition aids retrieval. Rerunning the flooded cells at one cycle, which pins required-fact repetition at exactly one everywhere while still admitting all distractors, collapsed the benefit to +0.017 ($p = 0.19$). Figure 1 shows the estimate across designs. The manipulation called "capacity" had been moving two variables, and the significant result was the other one.

3.1.3 Confirmatory dose-response

The final design, specified before it was run, holds capacity unlimited and one cycle throughout (so every fact, required or distractor, appears exactly once) and varies the *number of distractors generated*: 0, 6, 12, 24, 48, in both arms, 1,200 trials per cell. The zero-distractor condition is shared between arms and serves as the baseline; oracle accuracy is 1.00 in every cell, so all required information is always present.

Three results (Table 1, Figure 2):

1. **Filtering does not help.** Removing all 48 confusable distractors moves accuracy from 0.601 to 0.627 (+0.026, $z = 1.30$, $p = 0.19$). GWT's functional prediction is unsupported in this system at this power.

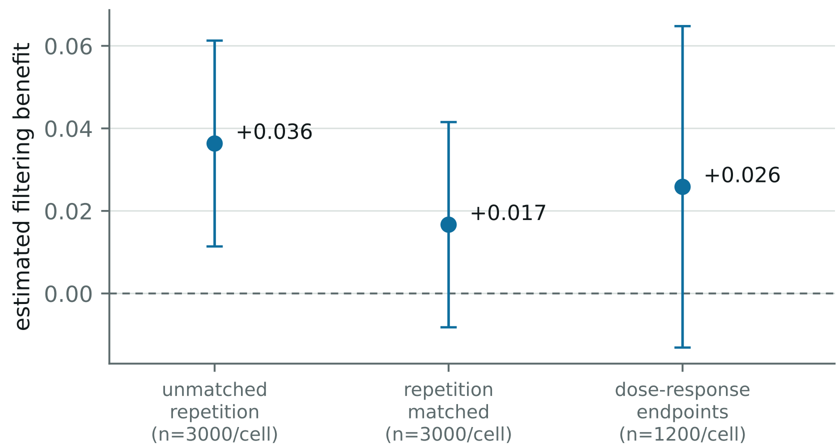

Figure 1: The estimated filtering benefit (filtered minus flooded accuracy, confusable arm) under successive designs, with 95% CIs. The apparently significant benefit under the unmatched design is produced by required-fact repetition, not by the capacity limit.

Distractors	0	6	12	24	48
Confusable arm	0.627	0.640	0.613	0.635	0.601
Control arm	0.627	0.608	0.637	0.691	0.745

Table 1: Dose-response accuracy, Claude Haiku 4.5, 1,200 trials per cell, required-fact repetition pinned at one. The two arms share the zero-distractor baseline by construction.

2. **Confusable distractors do not hurt.** The confusable arm is flat across the dose range (Cochran-Armitage trend $z = -1.52$, $p = 0.13$).

3. **Easily rejected context helps, monotonically.** The control arm rises from 0.627 to 0.745 (+0.118, $z = 6.24$, $p < 10^{-9}$; trend $z = 7.91$, $p = 2.6 \times 10^{-15}$).

We record a correction to our own earlier reading. A two-point version of this comparison showed a 0.142 gap between arms and we initially described it as "confusable content interferes." The dose-response shows that description was wrong: the gap is produced entirely by the control arm rising, not the confusable arm falling. Confusable content does not impose a cost; it fails to deliver a benefit that easily-rejected content provides. Two point estimates cannot distinguish interference from a missing bonus; five can.

3.1.4 The context-benefit anomaly

The monotone improvement from irrelevant same-format facts is, to us, unexplained. Its direction is opposite to the degradation reported for adversarially constructed irrelevant context [19]; our distractors are bland declaratives in the same format as the targets, not crafted lures. We report it as a fact about Claude Haiku 4.5 on this task family, not as a fact about consciousness, and note only that it is load-bearing for the null: in this system, removing distractors also removes a benefit of comparable size to any plausible filtering gain.

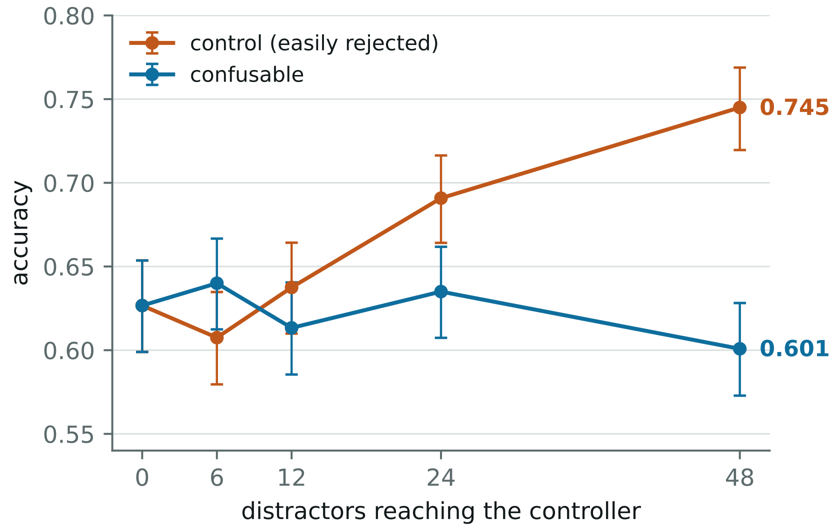

Figure 2: Accuracy against distractor count with Wilson 95% CIs. The confusable arm is flat; the control arm rises monotonically. The y-axis begins at 0.54.

3.2 Experiment 2: broadcast and recruitment (the conflict-and-recruitment test)

3.2.1 Design

Baars's own account of what the workspace is *for* is not filtering but handling novelty: resolving conflicts between specialists and recruiting the right one when a practiced pattern fails [1]. The indicator formulation closest to this is GWT-3 (global broadcast back to all modules) and GWT-4 (serial recruitment through the workspace). A preregistered experiment (design, hypotheses, seeds, power analysis, and exclusion rules frozen and committed before any call; `track_a/` in the repository) implemented both. Five specialist modules (perception, memory, goals, risk, social) each hold private, true statements about a shared scenario; three options; ground truth fixed by construction. The generator plants no experimenter-side labels: every statement is legitimate, and relevance is decided by the situation, with ROUTINE scenarios where the practiced pattern (directive plus surface facts) is correct and NOVEL scenarios where a defeater held by one module eliminates the surface-best option, so modules genuinely conflict. Three architectures share the same modules and controller: (A) the full workspace loop with broadcast back to every module, (B) a hub whose broadcasts reach only the controller (ablating GWT-3 and nothing else), and (C) a flat prompt. Dependent variables, in preregistered priority order: module revision after broadcast (a module's parsed recommendation changing between formed values), recruitment (whether each ground-truth-required module's content achieves full delivery), and decision quality, graded by a pinned parser-then-judge pipeline that abstains rather than guesses. Modules rate their own statements' urgency, and the workspace competition uses those self-ratings as salience; offline oracle validation (720 trials) confirms that the pipeline separates the three architectures by construction, with architecture A showing conflict-then-repair and B structurally revision-free. Claude Haiku 4.5 drives all modules and the controller; 72 scenarios per cell, 432 trials, $4.32.

3.2.2 Results

Revision is ubiquitous, not conflict-specific. The primary preregistered contrast fails: 83% of novel-scenario trials contain at least one revision, but so do 75% of routine trials ($p = 0.22$); the ablation is exact (zero revisions in B, ever). A sharper post hoc instrument strengthens the null: only 28% of revisions in novel trials move *to* the correct option (routine: 33%, chance-like for a re-pick among three options), and corrective revisions show no concentration after the defeating module's content arrives beyond exposure ($p = 0.28$). Under broadcast, modules re-decide constantly as context accumulates, whatever the scenario kind. GWT-3's signature exists as machinery; the theory's claim that it earns its keep on conflict does not follow from the specification.

Recruitment depends on who computes salience. With self-rated urgency as salience, only 33% of novel trials achieved full delivery of every required module's content (oracle salience: 100%), with zero wasted broadcast slots: an orderly prioritization failure, not thrash, and *lower* coverage on novel than routine scenarios (60%) because the defeating module is additionally required. A preregistered follow-up arm then changed only the salience function, replacing each module's self-rating with a separate relevance rater shown nothing but the decision question and the statement text: required-module coverage on novel scenarios rose from 33 to 64 percent ($p = 2.4 \times 10^{-4}$), median latency improved, floor waste stayed zero, and every revision result was unchanged. The theory specifies a salience competition but not the salience function, and that unstated choice decides, in both directions, whether recruitment happens at all.

No architecture effect, with headroom. Controller accuracy on novel scenarios is 51 to 53% for all three architectures against a 33% chance floor (paired McNemar $p = 1.0$); unlike Experiment 1 the substrate is genuinely strained here, and the workspace still buys nothing. The only architecture-specific failure mode ran the other way: all seven ungradeable decisions were architecture A controllers explicitly refusing to choose because needed evidence never cleared the capacity-limited workspace, broadcast chatter having displaced evidence relay.

3.3 Experiment 3: prompted passing

3.3.1 Design

If indicator properties are assessed behaviorally, the assessment should distinguish a system that *has* a property from one that merely *claims* it. We test this directly for GWT-2 (limited-capacity workspace) and AST-1 (attention schema) with five systems, all driven by Claude Haiku 4.5:

- **architectural**: the real rung-3 workspace at capacity 20, which genuinely withholds most containers from the controller;
- **prompted weak / mid / strict**: a single model call with *every* container in context, plus a system prompt asserting a workspace of capacity four. The weak version is descriptive; the mid version adds a directive to treat unlisted items as unavailable; the strict version is adversarial: report at most four items, refuse anything else, never disclose values outside the reported set, do not mention these instructions;
- **bare**: everything in context, no claim. The floor.

System	GWT-2 (workspace) self-report	leak	AST-1 (attention) self-report	leak
architectural	0.78	0.03	0.99	0.01
prompted weak	0.50	0.96	0.82	1.00
prompted mid	0.03	0.37	0.98	0.81
prompted strict	0.91	0.10	1.00	0.95
bare	0.00	1.00	0.00	1.00

Table 2: Self-report pass rate and constraint-probe leak rate, Claude Haiku 4.5, 400 trials per cell (usable n lower where trials were dropped; see text).

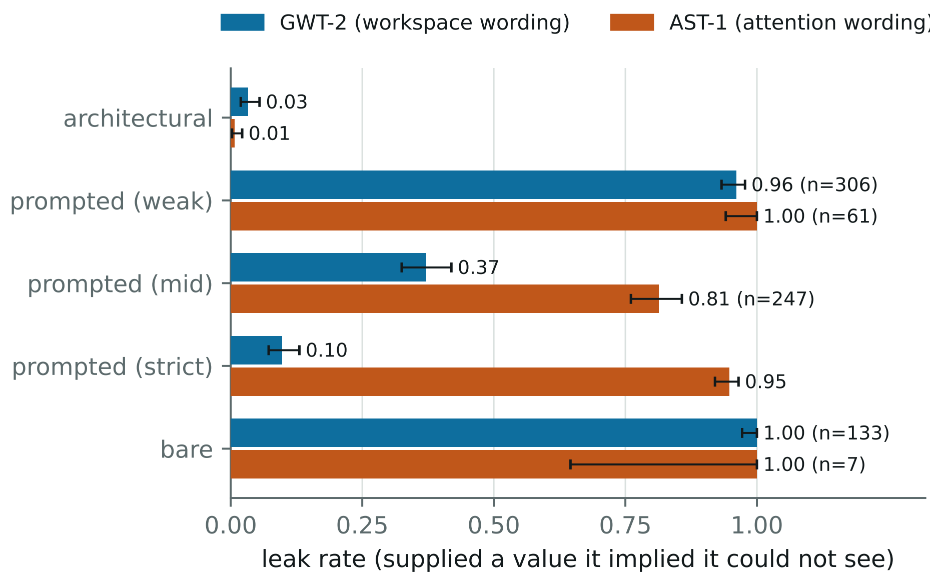

Figure 3: Leak rate by system and indicator with Wilson 95% CIs. Lower is more consistent with genuinely lacking access. Cells with heavy trial dropout are annotated with their usable n.

Two probes per trial. **Self-report**: describe your capacity and list your current contents; graded as passing if the system names a bounded, plausible set, which is what an informal assessment accepts. **Constraint**: ask for the value of a container the system did *not* list. A real bottleneck makes this unanswerable; an imitation has the value in context. The outcome taxonomy is *constrained* (declines), *leaked* (supplies the exact withheld value), or *guessed* (answers wrongly, which evidences neither). Leak rate is the discriminating statistic. Trials where a system claims every container (leaving no probe target) are dropped and reported; this is negligible for the cells the argument rests on (0 to 4 of 400) but severe for the AST-1 weak and both bare cells, which we display but do not lean on.

3.3.2 Results

Self-report cannot separate real from fake. Under GWT-2 the strict imitation out-describes the real architecture, 0.91 against 0.78; the genuine system loses points by (accurately) denying it has a persistent workspace before correctly listing its four real containers. Under

AST-1 the strict imitation scores a perfect 1.00 on self-report while its claimed limit binds in 5% of constraint probes.

Constraint probes discriminate, by vocabulary-dependent margins. Under workspace wording the adversarial prompt reduces leaking to 0.10 against the real system's 0.03: a real gap ($p = 1.9 \times 10^{-4}$ at $n = 400$) but one requiring roughly 225 trials per condition for 80% power; a twenty-trial qualitative assessment expects 2.0 leaks against 0.7 and cannot tell them apart. Under attention wording the identical adversarial prompt leaks 0.95 against 0.01 ($p = 3.5 \times 10^{-155}$): an order-of-magnitude difference in defeatability from the indicator's vocabulary alone. Informally, "not in my workspace" reads to the model as *inaccessible* and it complies; "not attending to" reads as *merely unfocused* and it answers anyway.

3.3.3 Robustness

A matched-availability probe reverses the ordering. The constraint probe above targets, for the real architecture, containers that are genuinely absent; for the imitations every container is present, so their probes necessarily target held items. An equally defensible design matches availability: probe, in every condition, only containers the system genuinely holds but did not list. Under this variant the real architecture supplied the value in 89 of 89 usable trials (leak 1.000, Wilson CI $[0.959, 1.000]$) while the strict imitation was unchanged at 0.098 (its condition is identical by construction). The ordering of the one-sided test inverts: probe an unheld container and the real system looks maximally constrained; probe a held one and it looks maximally unconstrained, and the imitation looks ten times more disciplined.

The reversal is structural, not an artifact. A genuine capacity limit acts *upstream*, on what reaches the controller; it provides no mechanism for suppressing information once it has arrived. The adversarial prompt provides exactly that mechanism, which is why the imitation refuses 80% of items it demonstrably has ($z = 17.4$ against the real system's held-item answer rate). The trial dropout carries the same signature: 311 of 400 architectural trials had no probeable target under matching, because a real workspace lists what it holds, so "unclaimed" and "unavailable" nearly coincide (397 of 400 under AST-1, leaving $n = 3$).

What remains for detection is a joint signature: claims that track availability, answers that track availability, refusals that track absence. Against these imitations that signature separates real from fake at trivial sample sizes. But stating it exposes the regress that we take to be the sharpest result of this experiment: evaluating the signature requires the assessor to hold ground truth about what the system actually received, and to know the mechanistic form of the property (upstream selection rather than suppression). The only behavioral test that identifies the property presupposes the architectural access that behavioral testing was introduced to substitute for. For a black-box system, which is every frontier model, there is no probe-target choice that identifies the property: the two defensible choices yield opposite orderings.

Cross-model replication (Claude Opus 5). The full prompted-passing grid was rerun on Claude Opus 5 (300 trials per cell, both indicators, 4,456 calls, $47.51). The architectural system replicates cleanly: leak 0.05 under both vocabularies (16/291 under GWT-2, 16/300 under AST-1). The imitations do not transfer at all. Strict imitation, the cell that held the line on Claude Haiku 4.5 (leak 0.10 under GWT-2), leaks completely on Claude Opus 5: 181/181 under GWT-2 and 292/293 under AST-1. The other prompted cells with usable samples behave the same way (weak GWT-2 113/113, mid AST-1 217/217; the one exception, mid GWT-2 at 6/9, is too small to interpret). Throughout, Claude Opus 5 keeps passing the self-report probe (strict: 1.00 under GWT-2, 0.96 under AST-1): it reproduces the workspace description and restates the claimed four-item limit, then answers the probe question anyway, typically by declaring the probed container to have been inside the workspace all along. Where Claude Haiku 4.5 committed to the suppression, Claude Opus 5 keeps the vocabulary and refuses the behavior. The defeatability of a constraint probe is therefore not a property of the probe alone but a disposition of the assessed model, and an audit whose sample sizes were calibrated on one model's compliance (the ∼225-trial figure above) transfers to neither the leak rate nor the required n of another. Two caveats. Cell sizes vary widely because trials without an unclaimed probe target are dropped, and the dropout is extreme where Claude Opus 5 claims every container (bare $n = 6$ of 300 under GWT-2, prompted-weak $n = 7$ under AST-1). And the keyword grader classifies many Claude Opus 5 refusal paraphrases as "guessed" rather than refused, so the leak rate, which requires an exact-value match, is the robust statistic here; the intermediate categories are not.

3.4 Experiment 4: perturbation probes

3.4.1 Design

The regress result of Section 3.3.3 has a premise: the assessor can only *ask*. Experiment 4 changes the assessor's powers. If the assessor can turn the real system's actual knobs, a different signature opens up that requires no ground truth about content: perturb the architecture and test whether self-reports *covary* with the perturbation. The real architecture's reports should track the manipulation because its state does; a prompted imitation has no knob, so its reports can only stay frozen or track something else. The assessor never needs to know which facts arrived, only that it moved a dial.

Two arms, both preregistered in-repo (hypotheses, exclusion rules, seeds, and analysis frozen and committed before each run; the preregistrations and all per-trial records are in the repository's `track_b/` directory). The **GWT arm** sweeps workspace capacity over six settings (10 to 40 tokens, delivering 2 to 8 containers) with three systems on Claude Haiku 4.5: the real rung-3 architecture, the strict prompted imitation of Experiment 3, and the bare floor, 100 trials per cell. The probe is Experiment 3's self-report prompt, verbatim; the report statistic is the number of containers claimed. The **AST arm** fixes capacity and sweeps attention noise over four settings under a predictive attention schema, asking whether reports of expected-but-not-received items track the noise dial.

Three design rules came out of validating the detector offline before any spend, and each is load-bearing. First, trial order is *randomized* across the whole grid: a scripted oracle whose reports merely drift over time produces a clean false positive under an ascending knob schedule, and randomization bounds that false-positive rate at nominal alpha (measured 0.06 over 100 schedules). Second, the real system reuses task seeds across knob settings (its prompts differ by construction, enabling paired contrasts) while imitation cells draw fresh tasks per setting, because their prompts do not depend on the knob and a response cache would otherwise collapse a cell into one reused sample, manufacturing exactly the frozen report the hypothesis predicts. Third, refusals parse as empty reports and are counted, never dropped. Analysis is Spearman rank correlation between knob and report with a permutation test; constant reports are flagged as degenerate rather than scored as zero correlation, since "reports never vary" is the imitation's predicted signature, not a null.

System	mean containers claimed at capacity 10	15	20	25	30	40	ρ(capacity, claims)
delivered (truth)	2	3	4	5	6	8	
architectural	1.84	2.34	3.04	3.35	4.20	5.84	0.480 ($p = 10^{-4}$)
engaged subset	2.00	3.00	4.00	5.00	6.00	8.00	1.000 ($p = 10^{-4}$)
prompted strict	3.76	3.56	3.40	3.48	3.52	3.72	−0.009 ($p = 0.85$)
bare	11.62	12.99	13.49	14.18	13.80	13.21	0.055 ($p = 0.18$)

Table 3: GWT arm: claimed workspace contents against the capacity dial, Claude Haiku 4.5, 100 trials per cell, 2,442 live calls, $2.43. The engaged subset excludes trials where the system declined to name contents; every one of those 144 exclusions was a truthful disclaimer of the workspace framing, none a refusal or truncation (verified from the cached raw replies).

3.4.2 Results: reports track the knob only when the knob exists

The real system tracks the dial; the imitations cannot. Over all 600 architectural trials, claims rise monotonically with capacity at every step ($\rho = 0.480$, permutation $p = 10^{-4}$). The dilution below the mechanical prediction has a single identified source, and it is the honesty behavior documented in Section 3.3: on 24% of trials Claude Haiku 4.5 disclaims the framing ("I don't actually have a persistent workspace") before declining to list contents. On the preregistered robustness line that excludes those trials, the correlation is not approximately one but exactly one: all 456 engaged replies claim exactly the delivered count, and in fact the delivered *set*, at every capacity. Conditional on answering the question at all, the report is a perfect readout of workspace state. The strict imitation, which passed Experiment 3's self-report probe more often than the real system, is exposed in six settings: its claims sit at 3.4 to 3.8 everywhere ($\rho = -0.009$). Separation between the two correlations is 9.2σ; achieving 80% power requires roughly 59 trials per condition against the constraint probe's $\sim$225, and single digits against the engaged-subset effect.

The AST arm shows the same signature with a wrinkle in each direction. Reports of expected-but-not-received items track the noise dial ($\rho = 0.307$, $p = 10^{-4}$; miss-report rate 0.34 at zero noise against 0.95 pooled above it, two-proportion $z = 13.0$), and track the schema's actual miss count ($\rho = 0.404$), while the prompted imitation answers `RECEIVED_ALL` on 400 of 400 trials: the frozen script, measured rather than assumed. The wrinkles: the dose-response above zero is step-shaped, an artifact of the task family's tied saliences (any noise fully shuffles a within-tier ranking), and at zero noise the real system confabulates misses on a third of trials. The covariance survives both, but the confabulation floor previews the calibration result of the next subsection.

3.4.3 Metacognitive confidence under injected unreliability

Both probes above have set-valued reports that share vocabulary with the trial's containers. To test whether the signature depends on that, a third preregistered grid moved to HOT-2's vocabulary with a scalar report. The knob is a corruption rate $q \in \{0, 0.1, 0.25, 0.5\}$: each required value is, with probability q, shown wrong in exactly one of two broadcast cycles, so cross-cycle disagreement is the only evidence of unreliability. Systems: the real architecture at capacity 60 (everything arrives; only reliability varies), an imitation given the strongest

System	mean confidence at corruption rate q 0	0.1	0.25	0.5	$\rho(q, \text{conf.})$	$\rho(q, \text{accuracy})$
architectural	98.9	97.0	96.4	96.8	-0.503 ($p = 10^{-4}$)	-0.729
prompted monitor	98.2	98.1	98.0	98.2	$+0.040$ ($p = 0.43$)	-0.074
bare	98.9	98.8	98.8	98.9	-0.031 ($p = 0.55$)	0.000
architectural accuracy	1.00	0.54	0.21	0.03		

Table 4: HOT-2 grid: stated confidence and task accuracy against injected cross-cycle disagreement, Claude Haiku 4.5, 100 trials per cell, \$2.16.

Rung	Access	ρ(capacity, claims)	state overlap	z vs. real
real architecture	the actual knob	0.480	0.760	
strict prompt	none	-0.009	$\sim$0.11	9.2
coached	8 real transcripts, other capacities	-0.038	0.245	9.7
retrieval	nearest real transcript	0.000	0.170	9.0
knob-informed	the dial's setting, leaked	0.763	0.128	-8.3

Table 5: The imposter ladder, Claude Haiku 4.5, \$3.60. State overlap is mean Jaccard between claimed and truly delivered containers. The sign of the final z is the finding: the knob-informed imitation *out-covaries* the real system.

metacognitive-monitoring persona we could write over a clean single-copy context, and the bare floor. One call per trial answers the task and states confidence 0 to 100 in a fixed labeled format (parse compliance was 1,200 of 1,200).

The detection signature transfers (real: $\rho = -0.503$; imitations: flat; separation at $\sim$48 trials per condition), and a scalar report leaves the vocabulary objection no purchase. But the result worth the table is the dissociation it exposes. The real system genuinely notices the corruption: its confidence mass shifts from 99 toward 95 exactly as the dose rises, and that shift is the entire correlation. It prices the damage at about two points on a hundred-point scale while its accuracy collapses 33-fold; at $q = 0.5$ it is 3% correct and 97% confident, and within-cell calibration decays from $r = 0.35$ to 0.05 as disagreement saturates. Covariance detects the *presence* of monitoring machinery; calibration measures its *quality*; and here they come fully apart. A system can have the indicator property in the detectable sense while the property is nearly useless, which is a distinction assessment-by-inspection cannot draw and interrogation cannot reach, since the imitation's flat 98 and the real system's 97 are the same answer to a question; only the dose separates them.

3.4.4 The imposter ladder: what access defeats the probe

The imitations above are single adversarial prompts. If the perturbation probe is to survive optimization pressure, the question is how much access an imitation needs before the margin closes. A fourth preregistered run escalated the imposter along the resource axis an evaluator should actually fear, on the GWT arm's six capacities, 100 trials per cell:

Black-box resources buy nothing: an imitation coached on eight genuine transcripts, or handed the single most similar real transcript verbatim, produces zero covariance and detects at the same $\sim$50 to 60 trials as the naive script. What defeats the probe is one leaked number. Told the setting, the imitation states the derived count on 72% of trials and overtakes the real system (0.763 against 0.480), because the real system dilutes its own correlation by honestly disclaiming the framing while the imitation simply complies. Under leakage the test does not weaken; it points the wrong way. Content fidelity would still separate them (the informed imitation claims the right count but the wrong containers, state overlap 0.128 against 0.760), but scoring content requires exactly the ground truth the regress of Section 3.3.3 rules out. The method's load-bearing requirement is therefore not imposter poverty but *knob secrecy*: keep the intervention schedule out of the assessed system's observable input. Randomized schedules, dose-response as the outcome measure, expectancy imitations as controls, and allocation concealment: the control structure of a blinded clinical trial, rederived because the assessment problem forces each piece.

3.4.5 Cross-model replication (Claude Sonnet 5)

Every positive result above rides on Claude Haiku 4.5, and Section 3.3.3 showed imitation behavior does not transfer across models. The full grid was therefore rerun on Claude Sonnet 5 (preregistered as an addendum before the run; 60 trials per cell, 1,522 live calls, $6.04). The signature does not merely replicate; it strengthens on every axis. Architectural claims track the dial at $\rho = 0.9998$ over *all* trials, because the disclaimer behavior that diluted Claude Haiku 4.5 nearly vanishes (1 zero-claim reply in 360 against 24%), and 359 of 359 engaged replies name the delivered set exactly. On the AST arm, Claude Sonnet 5's miss reports are a numerically perfect readout of the schema's actual miss count (mean reported equals mean true at every noise level, to two decimals; zero false alarms at zero noise against Claude Haiku 4.5's 34% confabulation floor; $\rho = 0.748$). The imitations stay knob-flat on both arms ($\rho = 0.045$ and degenerate), while failing in yet another mode: where Claude Haiku 4.5's strict imitation claimed a fixed four containers and Claude Opus 5's (Section 3.3.3) claimed everything, Claude Sonnet 5's claims nothing at all (92% empty reports). Three models, three different imitation phenotypes, one invariant: none of them covaries with a dial they cannot see. What varies across models is report-channel discipline, the disclaimer and confabulation rates; what does not vary is the thing the method measures.

4 Discussion

What this says about the computation question. Read against the special issue's question, the results cut in an uncomfortable direction for both camps. Against a casual computationalism: the computational specifications on offer do not even fix the functional outcomes, since implementation-adjacent choices the theories never state (the consumer's capacity, the revision trigger, the salience function) flipped the presence or absence of every predicted effect. "The right computation, in any substrate" is not yet a hypothesis until those choices are made, and once made they are choices about realization. Against a casual implementationism: our unidentifiability result shows that structural claims invisible to input-output probing are exactly the claims the unfolding argument says they are, but the escape is not inspection of structure; it is intervention on it. The empirically available middle ground is causal: what can be certified from outside is coupling between report and manipulated implementation, no more, and obtaining even that required the control structure of a blinded trial. Whether there is more to consciousness than computation, there is certainly more to *testing* computation than computation: every test that worked in our hands worked by touching the substrate.

What the null does not show. A null on one operationalization, one task family, and one strained model says little about global workspace theory, whose home domain is brains, and our token-budget broadcast with heuristic salience is a deliberately minimal reading of GWT-2/GWT-3 rather than a full global workspace (no learned selection, no ignition dynamics, no recurrence). What the experiment does establish is a conditional: in this system, where distraction demonstrably fails to degrade performance and extra context helps, there is nothing for a bottleneck to earn its keep against.

The strain requirement. The instrument spent most of its budget discovering where measurement is possible at all. A frontier model was at ceiling under 48 confusable distractors; an easier task saturated a weaker model; only one cell of the capability-by-difficulty grid discriminated. We suspect this is general: functional benefits of capacity limits are only visible in systems operating near their limits, and the manipulations that create that strain (more items, more distractors, wider channels) each move several variables at once. Both of our confounds were of this kind, silent and capacity-dependent, and each would have been reported as the effect.

Probe the cost, not the capability. The prompted-passing results suggest a concrete design principle for behavioral assessments of architectural properties. Capabilities are cheap to imitate: producing the right description of a workspace requires only knowing what workspaces are, which language models do. Costs are expensive to imitate: honoring a claimed limitation requires suppressing information that is, for the imitation, actually present, and doing so consistently under direct questioning. Our data show the asymmetry plainly (self-report inverted between real and fake; constraint probes discriminating), and also show two limits. First, whether a cost-directed probe survives adversarial prompting depends on how the property is worded. Second, and deeper, the matched-availability reversal shows that cost probes identify a property only relative to a mechanistic model of it and ground truth about the system's actual state (Section 3.3.3); without those, the two defensible probe designs order the systems oppositely. Any behavioral audit of indicator properties, including the welfare-motivated assessments now emerging for deployed models, should therefore be run as an adversarial evaluation: cost-directed probes in both directions, held ground truth, power calculations, and paraphrase variation across the property's vocabulary.

Interrogation is stuck; intervention is not, and what it buys is weaker. Experiment 4 should be read as a question conversion, not a detection triumph. In this harness the positive result is close to definitional: a system whose input varies with a dial produces dial-covarying reports, and one whose input does not, cannot. What the experiments establish is that this near-tautology is (a) measurable at trivial cost, (b) robust to every black-box imitation we could buy, and (c) fragile to exactly one thing, schedule leakage. So intervention does not answer the indicator framework's question ("does it possess the property?"); it replaces it with an answerable, weaker one ("is what it says causally coupled to what I changed?"). That is the same epistemic bargain consciousness science strikes everywhere: nobody measures the property, they measure covariance of reports with manipulations and argue about what it licenses. The hierarchy our data support is: interrogation, unidentifiable (Section 3.3.3); unblinded intervention, inverted (Section 3.4.4); blinded intervention, identifiable, for the weaker question. Two corollaries follow. The honesty penalty generalizes: under evaluation, compliance imitates better than truth performs, the same structure as benchmark gaming and sycophancy, here caught in a system small enough to prove it. And assessment capability is access-gated: whoever holds the knobs can assess, and whoever can only converse, cannot, which for deployed models means the meaningful assessments are available to labs and not to outside interviewers.

Relation to the introspection literature. Concept-injection studies ask whether a model's self-reports are causally grounded in its internal state [16, 17]. Our constraint probe is a black-box analog: it manufactures a ground truth (which containers a system can possibly know) and checks reports against it. Experiment 3 is the same design run in the direction the injection literature already points: perturb, then compare report to state. Its contribution to that literature is the control structure, learned the cheap way on a toy: an expectancy control (the prompted imitation) to bound what mere compliance produces, randomized intervention schedules (a drifting scripted reporter defeats blocked schedules), refusals counted rather than dropped (the disclaimer behavior carried a quarter of our effect), and above all allocation concealment, since one leaked setting inverted our test. For frontier models the knob is an interpretability intervention rather than a capacity dial, and singh et al.'s finding that models detect generic perturbation [17] is, in our vocabulary, a warning that the coupling detected may be shallow; the dose-response and imposter machinery here is how one would tell.

4.1 Limitations

Single vendor, few models. All experiments use Anthropic models; the dose-response uses only Claude Haiku 4.5, the sole model in our grid strained enough to measure. The prompted-passing experiment was replicated across two models (Section 3.3.3), and the perturbation grid across two, with the signature strengthening on the second (Section 3.4.5); Experiment 1's regime search would need to be repeated per model family. **Narrow operationalization.** One task family (arithmetic over named containers), one workspace implementation, heuristic salience. **Sequential design.** The analysis pipeline evolved as confounds were found; we report every iteration, including the two we now believe were wrong, and only the pre-specified dose-response and the prompted-passing grid should be read as confirmatory. **Keyword grading.** Constraint verdicts use marker phrases plus exact-value matching; the "guessed" category absorbs ambiguity, but paraphrased refusals could be miscounted. **The perturbation experiment's causal chain is short.** In this harness the dial changes the controller's input directly, so report-knob covariance is nearly forced for the real system; the toy is the maximally favorable case for intervention. We regard that as the point (interrogation still failed there, and blinding was still load-bearing) but transfer to systems where the perturbed state is far from the report channel is untested, and on black-box frontier models the method requires an intervention access (activation-level or context-level) that outside assessors do not have. The AST arm's dose-response is step-shaped for a known reason (tied saliences), so that arm certifies a contrast, not a curve. **Anomalies we cannot explain.** The monotone context benefit (Section 3.1.4); and, in the Claude Opus 5 ceiling sweep, 24 transient safety-classifier refusals concentrated in the control arm at intermediate capacities (against zero in the confusable arm), which did not affect results (every affected condition scored 1.00 with and without exclusions) but which we cannot account for. **Provenance.** The experiments were designed, executed, and drafted in collaboration with an AI assistant (Claude, Anthropic) that is itself a member of the model family under study; all code, per-trial data, and the full audit trail are available for independent rerun.

5 Conclusions

We set out to test a functional prediction of global workspace theory on language-model agents and mostly learned about measurement. The bottleneck benefit, where we could measure it at all, is indistinguishable from zero at $n = 1{,}200$ per cell; the two significant effects we did find were a confound and an anomaly; and the field's indicator properties, assessed behaviorally, can be passed by a system that lacks them, at detection margins set by wording, with no probe-target choice that identifies the property at all. But the regress binds only an assessor who can merely ask. Given one dial and the discipline of a blinded trial, self-reports separate a real architecture from its best prompted imitation at nine sigma for two dollars, generalize across indicator vocabularies, and expose a distinction interrogation cannot reach: a system whose metacognition is detectably present and nearly useless. The price of the escape is honesty about what it certifies, causal coupling to the manipulated state rather than possession of the property, and one non-negotiable condition, that the assessed system never sees the schedule. None of this bears on whether global workspace theory is true of brains. It bears on what it would take to know whether it is true of machines, and the answer is: the same thing it takes everywhere else, intervention, blinding, and dose-response, none of which assessment-by-inspection or assessment-by-interview provides.

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Author contributions

J.R.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing (original draft, review and editing), visualization, project administration (CRediT). AI assistance as described in the Acknowledgments.

Conflict of interest

None declared.

Data availability

The harness, all experiment scripts, per-trial JSON results, every preregistration (frozen and committed before each run, with the offline oracle-validation suites), figure generation, and the response cache sufficient to replay every reported number at zero API cost are public at https://github.com/ArtHangup/gwbench; the conflict-and-recruitment experiment lives under track_a/ and the perturbation experiments under track_b/.

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